

Homework (Carolina Reaper)

- Q1.** Solve for x in $|x| = 9$
(A) $x = 9$
(B) $x = -9$
(C) $x = 9$ or $x = -9$
(D) $x = 0$
(E) $x = 3$
- Q2.** What is the solution to $|2x - 5| = 11$?
(A) $x = 2$ or $x = 8$
(B) $x = -3$ or $x = 8$
(C) $x = 8$ or $x = -3$
(D) $x = -8$ or $x = 3$
(E) $x = -2$ or $x = -8$
- Q3.** A rhythm pattern has a frequency of $|x + 2| = 7$. What are the possible values for x ?
(A) $x = 5$ or $x = -9$
(B) $x = -5$ or $x = 9$
(C) $x = 3$ or $x = -11$
(D) $x = -3$ or $x = 11$
(E) $x = 9$ or $x = -5$
- Q4.** What is the solution to $|x + 4| - 3 = 9$?
(A) $x = 2$ or $x = -16$
(B) $x = -2$ or $x = 16$
(C) $x = 8$ or $x = -16$
(D) $x = 16$ or $x = -8$
(E) $x = -16$ or $x = 2$
- Q5.** Solve $|3x - 2| = |2x + 1|$
- (A) $x = 3$ or $x = -1$
(B) $x = -3$ or $x = 1$
(C) $x = 1$
(D) $x = -1$
(E) $x = 3$
- Q6.** A canvas has a size of $|x - 3| + 2 = 9$. What is x ?
(A) $x = 4$ or $x = 2$
(B) $x = 2$ or $x = 10$
(C) $x = 10$ or $x = 4$
(D) $x = 4$ or $x = -2$
(E) $x = 2$ or $x = -4$
- Q7.** Solve the equation $|x + 1| = |3x - 2|$
(A) $x = 1$
(B) $x = 1$ or $x =$
 $\frac{3}{2}$
(B) $x =$
 $\frac{3}{2}$
(B) $x = -1$ or $x =$
 $\frac{3}{2}$
(B) $x = -1$
- Q8.** The frequency of a sound wave is $|2x| = 12$. What is x ?
(A) $x = 6$ or $x = -6$
(B) $x = -6$ or $x = 6$
(C) $x = 6$
(D) $x = -6$
(E) $x = 0$

Open Ended Questions (FRQ)

- Q9.** Solve for x in $|x + 2| = |x - 5|$. Show all work and explain your reasoning.

- Q10.** A music composition has a rhythm pattern with $|x - 3| - 2 = 7$. Solve for x and explain how you would check for extraneous solutions.

Chef's Tips

- Read each question carefully before selecting an answer.
- Show all work and explain reasoning for free response questions.
- Check for extraneous solutions when solving absolute value equations.